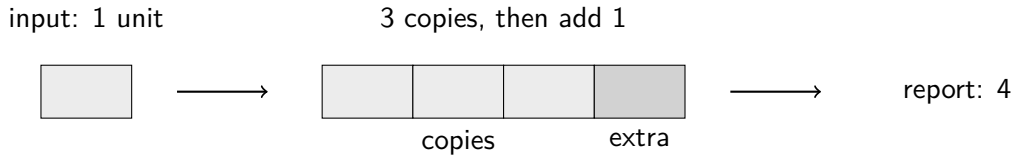
For Problems 1–2, a machine makes 1, 2, or 3 copies of the input length, then adds 0, 1, 2, 3, 4, 5, or 6 units. It keeps both choices fixed for the whole round. Each square in the paper row is one unit.




**Problem 1:** Play the machine. Secretly choose the number of copies and the extra amount. Your partner requests a strip's report. They may request more reports until they can name both choices. Reveal and switch jobs. How few requests can you manage?

| Round | Strips requested | Reports |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |



A

B

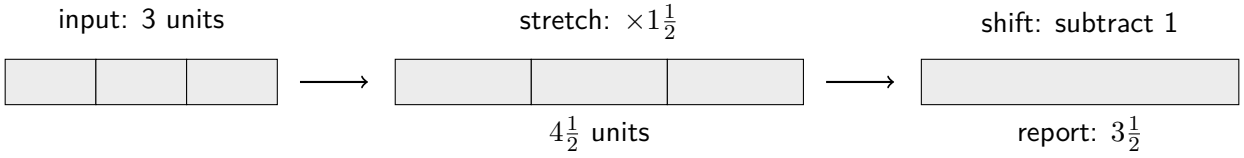
C

D

**Problem 2:** A 2-unit strip reports 6. Find every possible machine. Choose a second strip that would separate all your possibilities, and show the reports you might get. Could a single first request ever identify a machine?

| Copies | Extra amount | Second strip and its report |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

For Problems 3–4, a stretch-and-shift machine multiplies every input length by the same positive number, then adds the same fixed amount. Fractions are allowed; the added amount may be negative.




**Problem 3:** Each machine gives these two reports. Find its rule and its reports for lengths 0, 4, and 10. Could a different stretch-and-shift rule give the same two known reports?

| Machine A | Machine B |
|---|---|
| 2 units $\to$ 8<br>6 units $\to$ 16 | 2 units $\to$ 8<br>6 units $\to$ 12 |
| Rule: | Rule: |
| 0 units $\to$ | 0 units $\to$ |
| 4 units $\to$ | 4 units $\to$ |
| 10 units $\to$ | 10 units $\to$ |

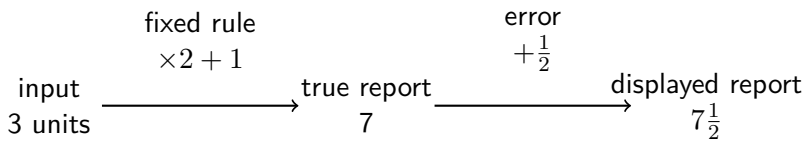
**Problem 4:** A machine claims to use one fixed stretch and shift. Could these three reports be correct? Find every way to repair the list by changing exactly one report.

| 2 units $\to$ 7 | 4 units $\to$ 11 | 6 units $\to$ 16 |
|---|---|---|

**Problem 5:** A machine gives 7, 11, and 15 for lengths 2, 4, and 6. Its rule stays fixed, but nobody promises stretch-and-shift. Make two complete report tables that fit these reports and disagree at length 8. Could any number of matching tests below length 8 force its report at 8?

| Length | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Table A | | 7 | | 11 | | 15 | | |
| Table B | | 7 | | 11 | | 15 | | |

For Problems 6–7, the machine uses one fixed stretch and shift. The reference lengths are exact. Each displayed report may be up to 1 above or below its true report.




**Problem 6:** For each pair of reference strips, find the smallest and largest possible stretch. Give a machine that reaches each extreme.

| Reference pair A | Reference pair B |
|---|---|
| 2 units: display 8<br>4 units: display 12 | 2 units: display 8<br>10 units: display 24 |

**Problem 7:** Choose two exact reference lengths from 1 to 10 units before seeing any reports. Which pair keeps the range of possible stretches narrowest, however unhelpful the errors are? Could repeating one reference ten times guarantee as narrow a range?